Research log: predicting BLEU from monolingual perplexity

lang-probing | `experiments/perplexity_bleu_linear/`

Saturday 23rd May, 2026

Purpose

This document is an append-only log of experiments studying how well monolingual FLORES perplexity predicts pairwise BLEU. Each section pins down one experiment: goal, method, code, data, results, takeaways. New experiments are added at the end.

The overarching hypothesis is **H1** in the project ledger: BLEU is predictable from monolingual source and target competence, with no cross-lingual interaction term needed. The rank-1 SVD of the BLEU matrix is 88% faithful for Llama-3.1-8B and 82% for Aya-23-8B (see `rank1_approximation.py`); the question is whether monolingual perplexity is a competent proxy for the implied per-language "competence" factors.

1 Setup (shared across experiments)

1.1 Models

- **Llama-3.1-8B** (`meta-llama/Meta-Llama-3.1-8B`, dense multilingual decoder, 32 transformer layers).
- **Aya-23-8B** (`CohereLabs/aya-23-8B`, Cohere's multilingual-instruction-tuned decoder of comparable size).

1.2 Data

All experiments here consume the joined metric file `data/pair_metrics.csv` (received from Jake, 2026-05-23; produced by `compile_shareable_metrics.py`). Two models $\times$ 24 languages $\times$ $24 \times 24 - 24 = 552$ directed pairs each (= 1,104 rows total). Schema is documented in `data/README.md`; the columns we use are `bleu`, `src_flores_perplexity`, `tgt_flores_perplexity`.

Coverage. Llama has perplexity for all 24 languages (552 pairs usable). Aya is missing FLORES perplexity for some languages, leaving 506 pairs usable for the linear analyses and 505 once we further require `bleu` > 0 for log models.

1.3 Where code lives

- `experiments/perplexity_bleu_linear/interaction_test.py` — Experiment 1.
- `experiments/perplexity_bleu_linear/rank1_latent_regression.py` — Experiment 2.
- `experiments/perplexity_bleu_linear/log_multiplicative.py` — Experiment 3.

All outputs go to `outputs/perplexity_bleu_linear/<experiment>/` and figures to `img/perplexity_bleu_linear/`.

2 Experiment 1: Does an src × tgt interaction term improve a linear-in-PPL fit?

2.1 Goal

The pre-existing linear fit (`run_linear_fit.py`) was a main-effects model. Before going more exotic, check whether a simple multiplicative interaction between source and target perplexity helps. If yes, the rank-1 faithfulness story might be reachable just by adding one term.

2.2 Method

For each model, fit two OLS models over all directed pairs:

$$\begin{aligned}
&\text{(a)} \quad \texttt{bleu} \sim 1 + \mathrm{PPL}_s + \mathrm{PPL}_t \\
&\text{(b)} \quad \texttt{bleu} \sim 1 + \mathrm{PPL}_s + \mathrm{PPL}_t + \mathrm{PPL}_s \cdot \mathrm{PPL}_t
\end{aligned}$$

Run a nested-model F-test: $F = ((\mathrm{RSS}_a - \mathrm{RSS}_b)/1)/(\mathrm{RSS}_b/(n-4))$ with $F \sim F(1, n-4)$ under H_0 (interaction coefficient is zero). The equivalent t-test on the interaction coefficient is also reported. Repeat with the perplexities log-transformed.

2.3 Results

model	transform	n	$R^2_{\text{no int.}}$	$R^2_{\text{+ int.}}$	ΔR^2	$F(1, \mathrm{df})$	p
llama	raw	552	0.0208	0.0210	0.00014	0.08	**0.78**
llama	log	552	0.0031	0.0033	0.00018	0.10	**0.75**
aya	raw	506	0.1048	0.1076	0.00281	1.58	**0.21**
aya	log	506	0.1048	0.1090	0.00423	2.38	**0.12**

Table 1: Nested-model F-test for the $\mathrm{PPL}_s \cdot \mathrm{PPL}_t$ interaction.

2.4 Takeaways

- The interaction term is **not statistically significant** for any (model, transform) combination at any conventional threshold.
- Llama remains in the $R^2 \approx 0.02$ regime; Aya is at ~ 0.10. Adding the interaction lifts R^2 by at most 0.4 percentage points.
- This rules out the simplest explanation for the 88% rank-1 faithfulness gap: a missing multiplicative term in PPL space.

Caveats.

- "For each language pair" would not be a fittable per-pair analysis (one observation per pair); we fit one model per language model instead.

- Aya's $n = 506$ vs Llama's 552 reflects missing Aya-side perplexities in the upstream data, not an inherent capability gap.

Outputs. `outputs/perplexity_bleu_linear/interaction_test/interaction_test.csv` `outputs/perplexity_bleu_linear/interaction_test/coefficients.csv`

3 Experiment 2: Rank-1 BLEU latents vs FLORES perplexity

3.1 Goal

The rank-1 SVD says BLEU $\approx u_s v_t$ for some per-language src-competence u_s and tgt-competence v_t. If FLORES perplexity is a faithful proxy for those latents, then *some* monotone transform of PPL should correlate strongly with u and v. This experiment tests that directly: recover u and v from the BLEU matrix, then correlate against PPL under several transforms.

If correlations are high but linear-in-PPL failed (Experiment 1), the problem is in *combining* src and tgt PPL, not in the per-language proxy. If correlations are low, FLORES PPL is not capturing competence in the rank-1 sense.

3.2 Method

1. Pivot `pair_metrics.csv` into a 24×24 BLEU matrix per model. Impute the NaN diagonal with column means (same convention as `rank1_approximation.py`).
2. SVD; extract rank-1 latents $u = U_{\cdot 1}\sqrt{\sigma_1},\ v = V_{\cdot 1}\sqrt{\sigma_1}$. Apply a sign convention so u correlates positively with row-mean BLEU.
3. For each language, collapse pairwise PPL to a per-language scalar (mean across pairs — in practice all values for a given lang are identical).
4. Compute Pearson and Spearman correlations between u_s and transforms $\{P, \log P, 1/P, 1/\sqrt{P}\}$ of source PPL, and similarly for v_t and target PPL.

3.3 Results

3.4 Takeaways

- **FLORES perplexity is not a faithful proxy for rank-1 BLEU competence**, especially for Llama. The strongest Spearman is $|\rho| = 0.43$ (Aya, src); for Llama, the strongest is $|\rho| = 0.31$ (src) and $|\rho| = 0.22$ (tgt). At $n = 24$, none of the Llama correlations are significant at $p < 0.05$.
- For Aya, $1/P$ on the source side is the best transform ($r = +0.41,\ p = 0.05$), suggesting some signal but not the dominant one. Spearman vs Pearson agreement is good — the relationship is monotone where it exists.
- This is the cleaner statement of the LEDGER's PER-vs-rank-1 tension: it is not that linear-in-PPL is the wrong functional form, it is that FLORES PPL barely encodes the per-language scalars that BLEU *does* factor through.
- The asymmetry (src signal > tgt signal in both models) is interesting and possibly diagnostic — target competence in translation is plausibly more about generation than about understanding perplexity-on-corpus.

model	role	transform	Pearson r	Pearson p	Spearman ρ
aya	src	raw	-0.312	0.147	-0.433
	src	log	-0.373	0.080	-0.433
	src	$1/P$	$+0.410$	**0.052**	$+0.433$
	src	$1/\sqrt{P}$	$+0.395$	0.062	$+0.433$
	tgt	raw	-0.368	0.084	-0.297
	tgt	log	-0.355	0.096	-0.297
	tgt	$1/P$	$+0.327$	0.128	$+0.297$
	tgt	$1/\sqrt{P}$	$+0.342$	0.110	$+0.297$
llama	src	raw	-0.222	0.296	-0.314
	src	log	-0.170	0.428	-0.314
	src	$1/P$	$+0.124$	0.565	$+0.314$
	src	$1/\sqrt{P}$	$+0.144$	0.502	$+0.314$
	tgt	raw	-0.148	0.492	-0.219
	tgt	log	-0.039	0.857	-0.219
	tgt	$1/P$	-0.076	0.723	$+0.219$
	tgt	$1/\sqrt{P}$	-0.022	0.920	$+0.219$

Table 2: Correlation between rank-1 BLEU latents and per-language FLORES perplexity transforms. $n = 24$ languages per model.

Outputs. `outputs/perplexity_bleu_linear/rank1_latent_regression/latent_vs_ppl_correlations.csv`
`outputs/perplexity_bleu_linear/rank1_latent_regression/latents_{llama,aya}.csv`
`img/perplexity_bleu_linear/rank1_latent_vs_ppl_{llama,aya}.png`

4 Experiment 3: Log-multiplicative fit

4.1 Goal

The rank-1 story is multiplicative ($\texttt{bleu} \approx u_s v_t$); the natural *linear* model in log space is therefore

$$\log \texttt{bleu} = a + b \log \mathrm{PPL}_s + c \log \mathrm{PPL}_t + \varepsilon, \quad \text{equivalently} \quad \texttt{bleu} = e^a \cdot \mathrm{PPL}_s^b \cdot \mathrm{PPL}_t^c.$$

Experiment 1 fit `bleu` directly with an interaction; this fits the right functional form for a multiplicative story. The question is how much this functional change buys.

4.2 Method

Drop pairs with $\texttt{bleu} \leq 0$ or missing PPL. Fit OLS in log space per model. Report:

- R^2 on the log scale (the scale we actually fit).
- R^2 on the BLEU scale (back-transformed predictions $\hat{y}_{\texttt{bleu}} = \exp(\hat{y}_{\log})$ vs actual BLEU). This is biased low by Jensen's inequality and is reported as a sanity check on practical utility, not as the model objective.
- F-test against intercept-only on the log scale.
- Per-coefficient t-tests.

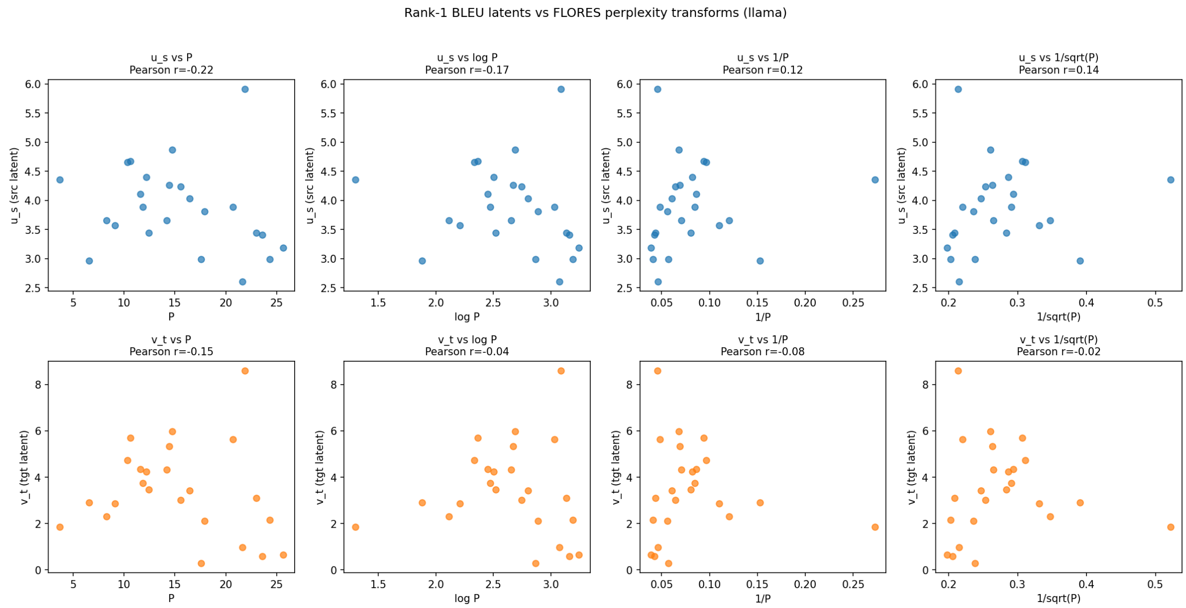

Figure 1: Llama: rank-1 latents vs FLORES PPL transforms.

model	n	$R^2_{\log}$	$R^2_{\text{BLEU bt}}$	$F(2, \text{df})$	p_F	coefs (a, b, c)
llama	544	0.0421	-0.196	11.89	8.9×10^{-6}	$(+3.48, -0.05, -0.38)$
aya	505	0.1357	-0.002	39.41	1.3×10^{-16}	$(+6.00, -0.39, -0.76)$

Table 3: Log-multiplicative fit. Coefficients b and c are the exponents on src and tgt PPL respectively in the back-transformed multiplicative form.

4.3 Results

Per-coefficient significance:

- **Llama:** $b_{\log P_s} = -0.05$ ($p = 0.53$, n.s.); $c_{\log P_t} = -0.38$ ($p = 1.6 \times 10^{-6}$, highly significant).
- **Aya:** $b_{\log P_s} = -0.39$ ($p = 5.0 \times 10^{-5}$); $c_{\log P_t} = -0.76$ ($p = 5.8 \times 10^{-15}$). Both significant.

4.4 Takeaways

- Switching to the right functional form (multiplicative in original scale, additive in log) does *help*: Llama moves from $R^2 = 0.02$ in raw space to $R^2_{\log} = 0.04$, and Aya moves from 0.10 to 0.14. The overall F-tests are highly significant.
- But: this is still nowhere near the rank-1 SVD's 88%/82%. Consistent with Experiment 2 — the issue is that $\log \text{PPL}$ is a weak monotone signal for the underlying competence factor, not that the combination form is wrong.
- **Target perplexity carries most of the signal** in both models. For Llama, source PPL is essentially uninformative once tgt PPL is in the model. The src/tgt asymmetry shows up again, mirroring Experiment 2's finding that tgt latents are even less well-explained by tgt PPL than src latents are by src PPL.

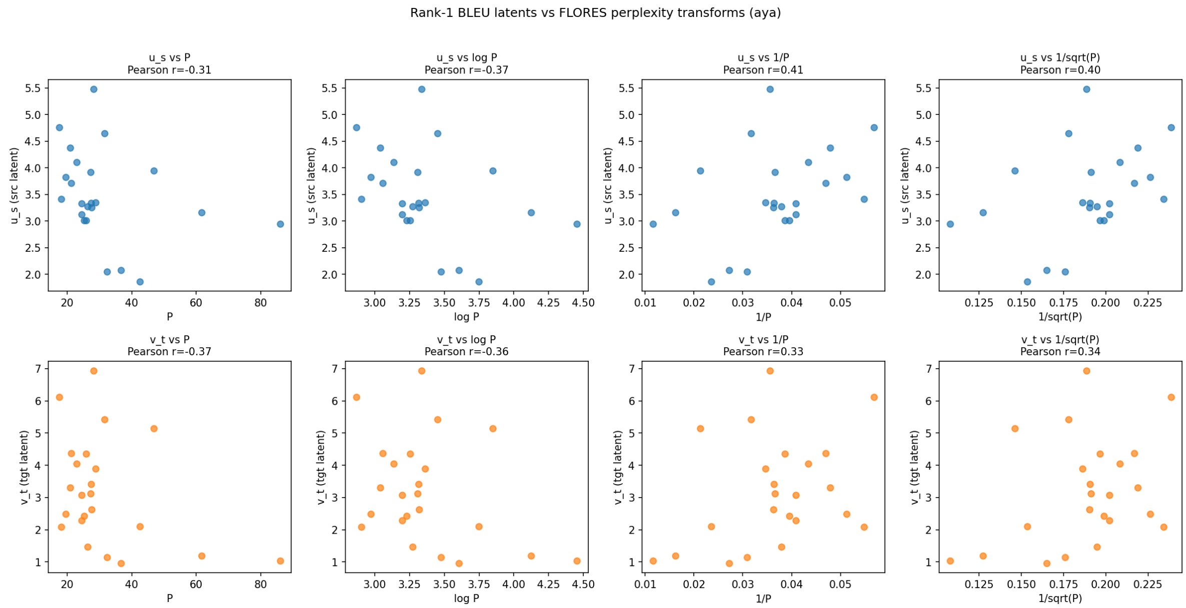

Figure 2: Aya: rank-1 latents vs FLORES PPL transforms.

- The negative back-transformed R^2 (especially Llama at -0.20) shows that the log-linear fit, when exponentiated, predicts BLEU *worse* than the BLEU mean. Don't read the model as BLEU-predictive in raw units; it is a likelihood model in log space.

Outputs. `outputs/perplexity_bleu_linear/log_multiplicative/fit_summary.csv`
`outputs/perplexity_bleu_linear/log_multiplicative/coefficients.csv`
`outputs/perplexity_bleu_linear/log_multiplicative/predictions_{llama,aya}.csv`
`img/perplexity_bleu_linear/log_multiplicative_{llama,aya}.png`

5 Interim status after Experiments 1–3

After three experiments, the empirical picture is:

1. Linear-in-PPL with or without an $\mathrm{PPL}_s \cdot \mathrm{PPL}_t$ interaction: $R^2 \leq 0.11$; interaction not significant (Experiment 1).

2. Rank-1 BLEU latents correlate only weakly ($|\rho| \leq 0.43$) with any tested monotone transform of FLORES PPL (Experiment 2).

3. Log-multiplicative model is highly significant overall but captures at most 14% of the log-BLEU variance (Experiment 3).

This motivates the next batch: re-cast competence as the per-language additive effects of a random-effects model, compare apples-to-apples against the rank-1 SVD on a centered scale, repeat the PPL correlations with language-labeled scatter plots, and unpack the src/tgt asymmetry that appears across Experiments 2 and 3.

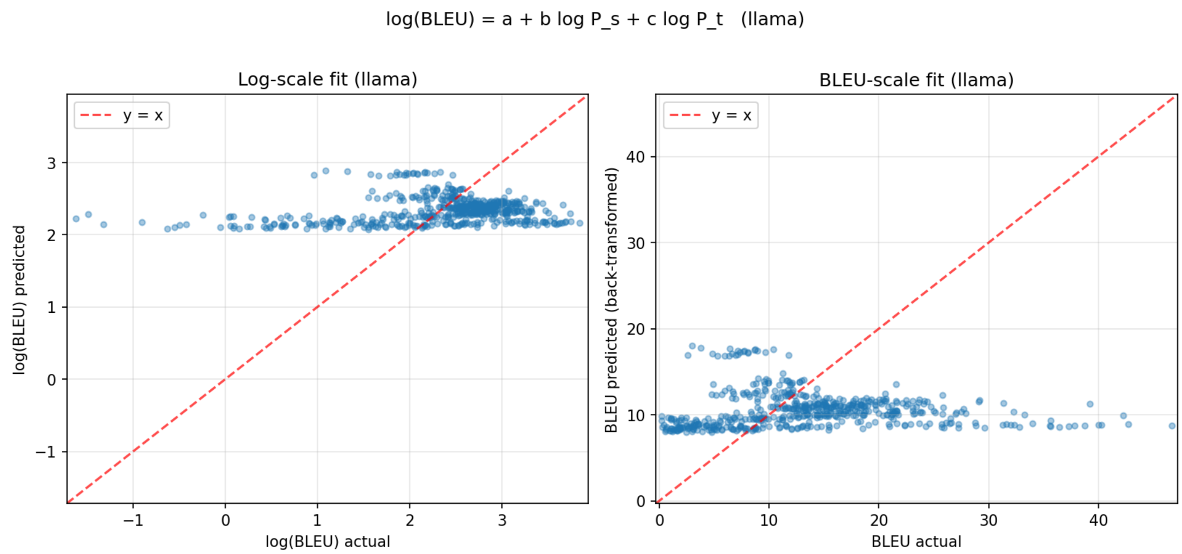

Figure 3: Llama: log-multiplicative fit, log-scale (left) and back-transformed (right).

6 Experiment 4: Additive RE model vs rank-1 SVD on a common scale

6.1 Goal

Fit the additive random-effects-style model

$$\texttt{bleu}_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij},$$

estimate per-language α (src effect) and β (tgt effect), and compare to rank-1 SVD on a comparable scale.

On RE-model vs PCA. The additive RE model is *not* identical to PCA/rank-1 SVD. RE is *additive*; rank-1 SVD is *multiplicative*. They coincide only up to a log transform: additive on $\log \texttt{bleu} \approx$ rank-1 SVD on `bleu`. Additive on raw BLEU is a separate model class.

A subtle but important detail: the original "rank-1 faithfulness = 88%" metric is

$$1 - \|M - M_1\|_F / \|M\|_F,$$

measured against the *uncentered* matrix norm. For Llama, $\sim 72\%$ of $\|M\|_F^2$ is the squared grand mean ($552 \cdot 13.27^2 \approx 97{,}166$ out of $\|M\|_F^2 \approx 135{,}297$), so the headline number is partly inflated by the metric rather than reflecting only structural fit. We report standard centered R^2 alongside it, and Section 9 gives the apples-to-apples structural comparison between additive and multiplicative model classes.

6.2 Method

Fit by OLS in long format with `src` and `tgt` dummies (drop the last category on each side for identifiability; re-center coefficients to sum to zero so μ is the grand mean and α, β are deviations). Then evaluate four decompositions on the observed cells using $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}_{\mathrm{centered}}$:

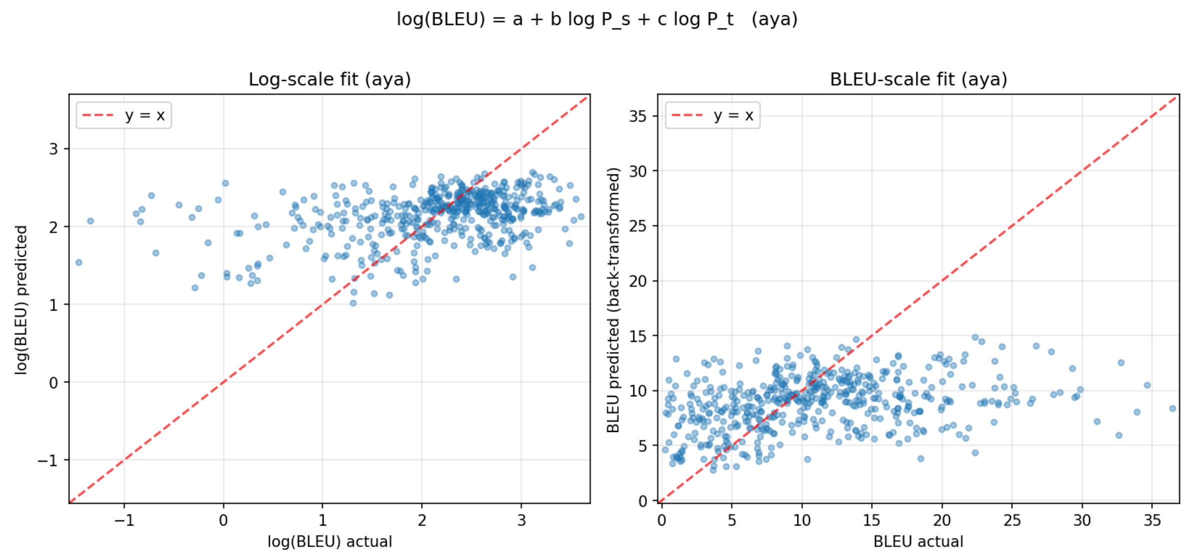

Figure 4: Aya: log-multiplicative fit, log-scale (left) and back-transformed (right).

- Additive RE (47 parameters: $1 + 2(n_{\text{lang}} - 1)$).
- Rank-1 SVD on uncentered M (existing approach).
- Rank-1 SVD on centered M (matches additive's centering).
- AMMI: additive + rank-1 of the additive residuals.

6.3 Results

model	$\bar{y}$	SD	add	SVD(M)	$\bar{y}$+SVD($M - \bar{y}$)	AMMI	faith. SVD
llama	13.27	8.32	0.932	0.960	0.859	0.965	0.894
aya	10.81	6.77	0.879	0.899	0.716	0.925	0.832

Table 4: First four model columns are standard centered $R^2 = 1 - \sum(y - \hat{y})^2 / \sum(y - \bar{y})^2$, on $n = 552$ observed cells per model. The final "faith. SVD" column is the original uncentered $1 - \|M - M_1\|_F / \|M\|_F$ metric. The two SVD-based R² columns differ substantively: column 4 ("SVD(M)") is the rank-1 SVD of the raw matrix used as a predictor of BLEU; column 5 ("$\bar{y}$+SVD($M - \bar{y}$)") is the structurally honest comparison to the additive model – both use the grand mean as a constant offset and the rank-1 captures only the centered residual. See Section 9.

6.4 Takeaways

- The 88% headline "faithfulness" is on an uncentered Frobenius scale where the squared grand mean dominates $\|M\|_F^2$ (Llama: 72%; Aya: 66%). On standard centered R^2, the same rank-1 SVD of uncentered M achieves 0.960/0.899 (Llama / Aya); the headline is not "mostly the mean", but the metric inflates it.

- The four R^2 columns measure different things. Comparing additive (0.932/0.879) to "SVD(M)" (0.960/0.899) is *not* the structural "additive vs multiplicative" question, because the single rank-1 outer product is allowed to absorb the mean, the per-language main effects, *and* multiplicative interaction all together. The apples-to-apples structural comparison is additive vs. "$\bar{y}$+SVD($M - \bar{y}$)" – both with the grand mean as a constant offset, both estimating only per-language structure. That comparison and its interpretation live in Section 9.
- AMMI (additive + rank-1 of additive residuals) lifts R^2 to 0.965/0.925, only 3–5 percentage points above additive. The genuinely multiplicative residual interaction in BLEU is small.

Outputs. `outputs/perplexity_bleu_linear/additive_random_effects/fit_summary.csv`
`outputs/perplexity_bleu_linear/additive_random_effects/coefficients_{llama,aya}.csv`
`outputs/perplexity_bleu_linear/additive_random_effects/bleu_matrix_{llama,aya}.csv`

7 Experiment 5: Additive α, β effects vs FLORES PPL

7.1 Goal

Repeat Experiment 2's correlation study with the additive-model coefficients in place of the SVD vectors. The additive coefficients are more directly interpretable as "how much above/below the grand-mean BLEU does this language tend to contribute as src / tgt". Produce language-labeled scatter plots to identify outlier languages.

7.2 Results

model	role	transform	Pearson r	Pearson p	Spearman ρ
aya	src (α)	raw	−0.304	0.158	−0.418
	src (α)	log	−0.362	0.089	−0.418
	src (α)	$1/P$	+0.398	0.060	+0.418
	src (α)	$1/\sqrt{P}$	+0.384	0.071	+0.418
	tgt (β)	raw	−0.358	0.094	−0.292
	tgt (β)	log	−0.344	0.108	−0.292
	tgt (β)	$1/P$	+0.316	0.142	+0.292
	tgt (β)	$1/\sqrt{P}$	+0.331	0.123	+0.292
llama	src (α)	raw	−0.188	0.378	−0.306
	src (α)	log	−0.141	0.511	−0.306
	src (α)	$1/P$	+0.104	0.630	+0.306
	src (α)	$1/\sqrt{P}$	+0.119	0.579	+0.306
	tgt (β)	raw	−0.145	0.500	−0.216
	tgt (β)	log	−0.037	0.862	−0.216
	tgt (β)	$1/P$	−0.076	0.724	+0.216
	tgt (β)	$1/\sqrt{P}$	−0.022	0.917	+0.216

Table 5: Correlation between additive-model α, β and per-language FLORES PPL. $n = 24$ per row. Compare to Experiment 2: numbers are nearly identical, confirming that the additive coefficients and the rank-1 SVD latents are estimating closely related per-language scalars (after subtraction of the mean, which dominates the SVD).

7.3 Outlier languages (regress α or β on $\log P$, take largest signed residuals)

- **Llama, src (α):** `eng` (+8.47), `kor` (−4.37), `hin` (−4.28). English is far above its PPL-predicted src competence; Korean and Hindi are below.
- **Llama, tgt (β):** `eng` (+21.07), `jpn` (−12.15), `zho_simpl` (−10.68). English is a massive positive outlier as a target; Japanese and Chinese are massive negative outliers — much lower BLEU as target than their FLORES PPL would predict. Tokenization on CJK is a known confound for BLEU.
- **Aya, src (α):** `eng` (+7.62), `jpn` (−4.79), `tur` (−4.14).
- **Aya, tgt (β):** `eng` (+13.46), `ind` (+9.58), `por` (+8.33). All three are positive outliers; Indonesian and Portuguese "over-deliver" as targets relative to their PPL.

Implication. The PPL-vs-competence story is dominated by a small number of outlier languages, especially English (always a positive outlier in magnitude) and CJK targets in Llama (always a negative outlier). Removing English alone would likely change the headline correlations substantially.

Outputs. `outputs/perplexity_bleu_linear/alpha_beta_vs_ppl/correlations.csv` `img/perplexity_bleu_linear/alpha_beta_vs_ppl_{llama,aya}.png`

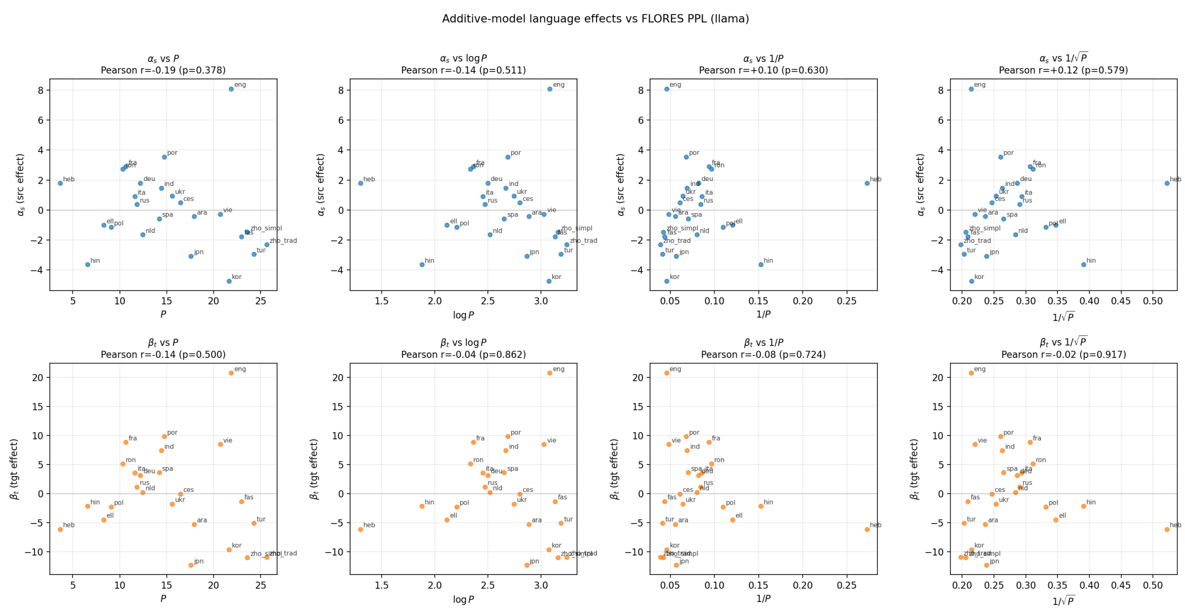

Figure 5: Llama: additive α_s and β_t vs FLORES PPL transforms, labeled by language. English is a positive outlier; `jpn` and `zho_simpl` are far-negative as targets.

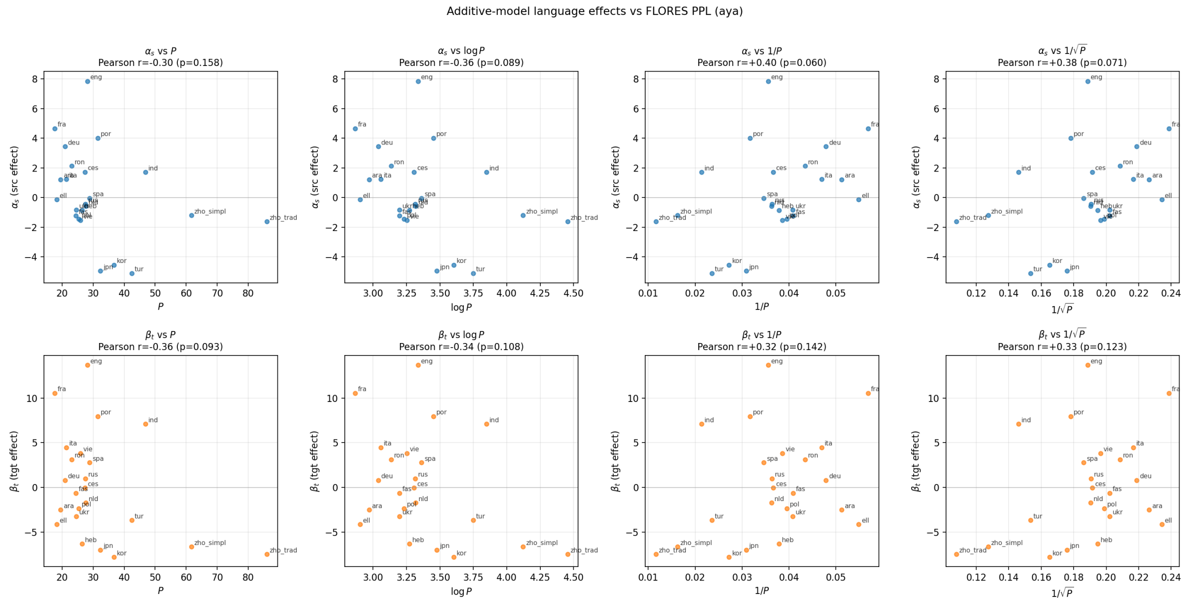

Figure 6: Aya: additive α_s and β_t vs FLORES PPL transforms, labeled by language.

8 Experiment 6: Why src signal > tgt in Experiments 2/5 but tgt signal > src in Experiment 3?

8.1 The puzzle

In Experiments 2 and 5 (per-language: $n = 24$) the source-side Spearman correlation with PPL is stronger than the target-side: $|\rho_{\text{src}}| > |\rho_{\text{tgt}}|$ for both models. In Experiment 3 (per-pair: $n \approx 550$) the target-side log-PPL coefficient is much larger in absolute value than the source-side. These look like opposing claims; they are not.

8.2 Diagnostic results

D1 — Variance decomposition of BLEU across languages.

model	var(mean BLEU as src)	var(mean BLEU as tgt)	ratio tgt/src
llama	6.11	56.62	**9.26**
aya	7.66	32.69	**4.27**

Between-tgt-language BLEU variance is **4–9**× larger than between-src-language variance. Choosing a different target moves BLEU much more than choosing a different source.

D6 — Sequential ANOVA: how much R^2 does each side carry?

model	R^2 src only	R^2 tgt only	R^2 additive both	marg. gain src|tgt	marg. gain tgt|src
llama	0.0911	**0.8208**	0.9307	0.1099	**0.8396**
aya	0.1617	**0.6889**	0.8768	0.1880	**0.7151**

For Llama, tgt dummies *alone* explain 82% of BLEU variance; src dummies alone explain only 9%. This is the structural reason a per-pair regression places almost all the weight on tgt PPL: that is where almost all of the BLEU variance lives.

D3/D5 — Per-pair correlations. Per-pair univariate correlations on log scale:

model	$r(\log B, \log P_s)$	$r(\log B, \log P_t)$
llama	−0.018	−0.203
aya	−0.155	−0.327

Partial correlations (controlling for the other side):

model	$r(\log B, \log P_s \mid \log P_t)$	$r(\log B, \log P_t \mid \log P_s)$
llama	−0.027	−0.204
aya	−0.180	−0.338

Univariate and partial correlations are essentially equal — $\log P_s$ and $\log P_t$ are nearly uncorrelated in the per-pair design (as they must be, given a balanced full-grid of language pairs). So the "tgt-dominant" coefficient in Experiment 3 is not a multicollinearity artifact.

D2 — Per-language correlations of mean BLEU with PPL.

model	$r(\bar{B}_{\text{as-src}}, \log P_s)$	Spearman	$r(\bar{B}_{\text{as-tgt}}, \log P_t)$	Spearman
llama	−0.172	−0.327	−0.034	−0.216
aya	−0.359	−0.438	−0.342	−0.295

Once we collapse to per-language scalars, src and tgt look closer. Llama target is essentially uncorrelated with $\log P_t$ in Pearson (driven by the Japanese/Chinese outliers identified in Experiment 5).

8.3 Resolution

The two views measure orthogonal things:

- **Per-pair regression (Exp 3)** asks "how much BLEU variation does $\log P$ explain?" The answer is dominated by which side has more BLEU variation to explain. tgt has 4–9× the between-language variance of src (D1), so even at the same per-language correlation strength, tgt PPL has more to "soak up" and its regression coefficient is larger and more significant.
- **Per-language correlation (Exp 2 / 5)** asks "how well-ordered are the languages by PPL?" This is scale-invariant: it does not care how much BLEU variance is on each axis, only whether the *ranks* of languages line up. Here, tgt is hurt by a few large outliers (`jpn`, `zho_simpl` for Llama — see Experiment 5) that drag the correlation down, while src is more uniform.

So both observations are correct and compatible: **tgt is the high-leverage axis** for BLEU prediction, but **src has the cleaner per-language PPL signal** once we strip out scale and look only at ordering.

9 Experiment 7: Apples-to-apples additive vs multiplicative (subtracting the grand mean)

9.1 Why subtract the grand mean

The BLEU matrix has a large positive grand mean ($\bar{y} \approx 13$ for Llama, ≈ 11 for Aya). Any model with a free intercept or constant offset captures this trivially. The Frobenius faithfulness metric

$$1 - \|M - M_1\|_F / \|M\|_F$$

used in the original "88%" claim is measured against the uncentered $\|M\|_F$, where the squared grand mean accounts for $\sim 72\%$ of the denominator (Llama). So that headline metric does *not* cleanly distinguish "rank-1 captures the mean" from "rank-1 captures per-language structure".

To compare model classes *structurally* – additive main effects vs. multiplicative interaction – we put both on equal footing:

- Both models are given the grand mean as a fixed constant offset (not credited as "explained" by the model).
- Each model estimates per-language structure with a comparable parameter budget.
- Performance is measured on *centered* variance, $R^2 = 1 - \sum(y - \hat{y})^2 / \sum(y - \bar{y})^2$. The denominator already subtracts $\bar{y}$, so neither model is rewarded for matching it.

Equivalently: in OLS the additive model satisfies $\hat{\mu} = \bar{y}$ exactly, so fitting $\mu + \alpha + \beta$ to M is the same as fitting $\alpha + \beta$ to $(M - \bar{y})$. Doing the analogous mean subtraction for the rank-1 model gives the structurally honest comparison.

9.2 Model specifications

model	functional form	per-lang free parameters
additive (RE)	$\hat{y}_{ij} = \bar{y} + \alpha_i + \beta_j$	$2(n-1) = 46$
multiplicative	$\hat{y}_{ij} = \bar{y} + u_i v_j$	$2n - 1 = 47$
AMMI	$\hat{y}_{ij} = \bar{y} + \alpha_i + \beta_j + u_i v_j$	~ 93

For additive, α and β are sum-to-zero. For multiplicative, uv^T is the rank-1 SVD of $(M - \bar{y})$, then added back to $\bar{y}$ at prediction time. AMMI nests additive: the rank-1 of additive *residuals* is the additional term. Parameter budgets per language are essentially equal for additive and multiplicative (46 vs 47); AMMI has roughly twice as many.

9.3 Results

9.4 Interpretation

1. **Additive wins the structural comparison.** With essentially equal per-language parameter budgets and the grand mean given to both, the additive model explains substantially more centered BLEU variance than multiplicative: 0.93 vs 0.86 for Llama (+7 pp), 0.88 vs 0.72 for Aya (+16 pp). The advantage is larger for Aya, consistent with Aya's BLEU matrix being less dominated by a single axis: in Experiment 6 the between-language variance ratio tgt:src was 9.3 for Llama vs 4.3 for Aya. A rank-1 outer product can absorb one dominant axis well; when both src and tgt axes carry comparable structure, it cannot fit them both.

model	Llama R^2_{centered}	Aya R^2_{centered}
intercept only ($\hat{y} = \bar{y}$)	0.000	0.000
multiplicative ($\bar{y}$+rank-1 of $M - \bar{y}$)	0.859	0.716
additive ($\bar{y} + \alpha + \beta$)	**0.932**	**0.879**
AMMI (additive + rank-1 of additive residuals)	0.965	0.925

Table 6: Centered R^2 for like-for-like comparisons. Both single-structure models share the grand mean as a constant offset and have comparable parameter budgets (46 vs 47); they differ only in functional form.

2. **Multiplicative is not vacuous.** $R^2 = 0.86/0.72$ is far from zero. For Llama specifically, the centered BLEU matrix is dominated by target main effects (β), which a single outer product can approximate fairly well – this is why the multiplicative R^2 is high for Llama despite an additive matrix structure. The rank-1 fit is mostly capturing β projected through one outer product, not genuine multiplicative interaction.

3. **AMMI shows multiplicative residual interaction is small.** Adding rank-1 of the additive residuals on top of the additive model lifts R^2 only +3.3 pp for Llama ($0.932 \to 0.965$) and +4.6 pp for Aya ($0.879 \to 0.925$). This is the magnitude of *genuinely* multiplicative interaction, above and beyond additive main effects. It is real but small. Most of BLEU's per-language structure is additive; only a thin slice is multiplicative-on-top.

4. **Reframing the 88% headline.** The original "rank-1 SVD is 88% faithful" claim was on the uncentered Frobenius scale and is inflated by the matrix's grand mean. The structurally precise statement, on a like-for-like scale, is: *BLEU's per-language structure is dominantly additive (*93%/88%*), with a small multiplicative residual on top (*+3–5 pp*).* "H1 holds" in the sense that BLEU does factor cleanly into per-language source and target competence; the factorization is additive rather than multiplicative.

Note on rank-1 of uncentered M. For completeness: a model $\hat{y}_{ij} = (uv^T)_{ij}$ with no constant offset, fit by rank-1 SVD of the raw matrix M, reaches centered $R^2 = 0.960/0.899$ – slightly above additive. This is *not* a contradiction with point 1. That single outer product is doing triple duty: capturing mean structure, per-language deviations, and any interaction, all packed into one uv^T. It is a perfectly legitimate model, but it is not a structural "multiplicative" model in the sense of "deviation from the mean is $u \cdot v$" – it lets rank-1 implicitly absorb additive main-effects structure into its outer product. The like-for-like comparison above is the one that answers the structural question.

10 Experiment 8: Drop-English ablation — outlier-driven vs systematic PPL signal

10.1 Goal

Experiment 5 flagged English as the single largest positive outlier in both α (src effect) and β (tgt effect): $\alpha_{\text{eng}} = +8.5$ (Llama) / +7.6 (Aya), $\beta_{\text{eng}} = +21.1$ (Llama) / +13.5 (Aya) measured as signed residuals to the log-PPL trendline. With $n = 24$, a single high-leverage point can dominate Pearson correlations. The question: how much of Experiment 5's weak α, β vs PPL signal is English-driven, and how much is systematic across the other 23 languages?

10.2 Method

1. Drop all pairs where `src` = `eng` or `tgt` = `eng` from `pair_metrics.csv`: 552 → 506 pairs (23×22).
2. Refit the additive RE model on this reduced grid; recover new α and β over 23 languages.
3. Compute Pearson and Spearman correlations of new α, β against the same four PPL transforms used in Experiment 5 (`raw`, log, $1/P$, $1/\sqrt{P}$).
4. Compare side-by-side with the full 24-lang correlations.
5. Produce annotated scatter plots over the 23-lang set.

10.3 Structural fit unchanged

The additive R^2 is essentially identical after dropping English:

model	R^2 (24 langs)	R^2 (23 langs, no eng)	Δ
llama	0.932	0.927	−0.005
aya	0.879	0.862	−0.017

The grand mean drops slightly ($\bar{y}$ Llama 13.27 → 12.02, Aya 10.81 → 9.88) as expected when removing English's high-BLEU rows and columns, but the per-language additive structure still explains $\sim$ 90% of centered BLEU variance. So removing English doesn't change the structural story of Section 9.

10.4 PPL correlations: side-by-side comparison

10.5 Observations

- **Llama: English was suppressing most of the signal.** Pearson $|r|$ *roughly doubles* on both src and tgt after removing English. Llama src raw $|r|$ goes 0.19 → 0.44; Llama tgt raw $|r|$ goes 0.15 → 0.34. Llama src raw is now statistically significant ($p = 0.035$).
- **Aya: signal is systematic, not English-driven.** Pearson $|r|$ changes by only 0–8 pp after removing English. The correlation was already ~ 0.4 and stays at ~ 0.4. Aya src $1/P$ and $1/\sqrt{P}$ cross the $p < 0.05$ threshold, but the magnitude shift is small.
- **Spearman is robust either way.** Spearman shifts much less than Pearson when English is removed — the rank structure was less affected by the single outlier than the Pearson magnitudes were. The gap between Spearman (~ 0.45 src for both models) and Pearson on the full set therefore measures how much one outlier was distorting Pearson.
- **After removing English, the two models look similar.** Both have $|r|$ in the 0.3–0.45 range for the best transforms, with Aya slightly stronger. The original "Aya much better than Llama" verdict from Experiments 2/5 was, for Llama in particular, a Pearson-vs-outlier artifact.
- **Best transforms (no eng).** Llama src: `raw` ($r = -0.44$). Aya src: $1/P$ ($r = +0.48$). Both directions correct (higher PPL → lower α src effect). Transforms differ slightly between models but all are monotone negative in P.

model	transform	Pearson r full (24)	no eng (23)	$\Delta\|r\|$	Spearman ρ full	no eng	$\Delta\|\rho\|$
src (α)							
llama	raw	−0.188	**−0.441****	+0.252	−0.306	−0.453	+0.147
llama	log	−0.141	−0.357	+0.216	−0.306	−0.453	+0.147
llama	$1/P$	+0.104	+0.264	+0.160	+0.306	+0.453	+0.147
llama	$1/\sqrt{P}$	+0.119	+0.308	+0.189	+0.306	+0.453	+0.147
aya	raw	−0.304	−0.310	+0.006	−0.418	−0.444	+0.026
aya	log	−0.362	−0.404	+0.041	−0.418	−0.444	+0.026
aya	$1/P$	+0.398	**+0.476***	+0.078	+0.418	+0.444	+0.026
aya	$1/\sqrt{P}$	+0.384	**+0.444***	+0.060	+0.418	+0.444	+0.026
tgt (β)							
llama	raw	−0.145	−0.343	+0.198	−0.216	−0.308	+0.091
llama	log	−0.037	−0.189	+0.151	−0.216	−0.308	+0.091
llama	$1/P$	−0.076	+0.009	−0.067	+0.216	+0.308	+0.091
llama	$1/\sqrt{P}$	−0.022	+0.096	+0.074	+0.216	+0.308	+0.091
aya	raw	−0.358	−0.390	+0.032	−0.292	−0.340	+0.048
aya	log	−0.344	−0.389	+0.046	−0.292	−0.340	+0.048
aya	$1/P$	+0.316	+0.378	+0.062	+0.292	+0.340	+0.048
aya	$1/\sqrt{P}$	+0.331	+0.385	+0.054	+0.292	+0.340	+0.048

Table 7: Correlation of additive α, β with FLORES PPL transforms, full 24-lang set vs. 23-lang set after dropping English. $\Delta|r| = |r_{\text{no eng}}| - |r_{\text{full}}|$ (positive $\Rightarrow$ stronger correlation after removing English). Bold * marks $p < 0.05$ in the no-eng column.

10.6 Conclusion

The Experiment 5 narrative "FLORES PPL barely correlates with per-language competence" was substantially overstated for Llama. Once English (a high-leverage positive outlier on both src and tgt sides) is removed, the remaining 23 languages show systematic, moderate, and statistically detectable correlations in the expected direction ($|r| \approx 0.35$–0.45, $|\rho| \approx 0.31$–0.45).

The PPL signal is therefore best described as **moderate and systematic across non-English languages, with English as a strong positive outlier** that obscures the signal in the full sample. This is consistent with English being heavily over-represented in training data: it has unusually good translation quality *for its perplexity*, breaking the monotonic PPL$\rightarrow$competence relationship.

A natural follow-up: drop the next-largest outliers (`jpn`, `zho_simpl` for Llama tgt) and see if correlations strengthen further — or test with bootstrap resampling whether the remaining ~ 0.4 correlations would have been detectable at $n = 23$ regardless.

Outputs. `outputs/perplexity_bleu_linear/drop_english_ablation/correlations_no_eng.csv`
`outputs/perplexity_bleu_linear/drop_english_ablation/comparison_with_full.csv`
`outputs/perplexity_bleu_linear/drop_english_ablation/coefficients_no_eng_{llama,aya}.csv`
`outputs/perplexity_bleu_linear/drop_english_ablation/fit_summary_no_eng.csv`
`img/perplexity_bleu_linear/alpha_beta_vs_ppl_no_eng_{llama,aya}.png`

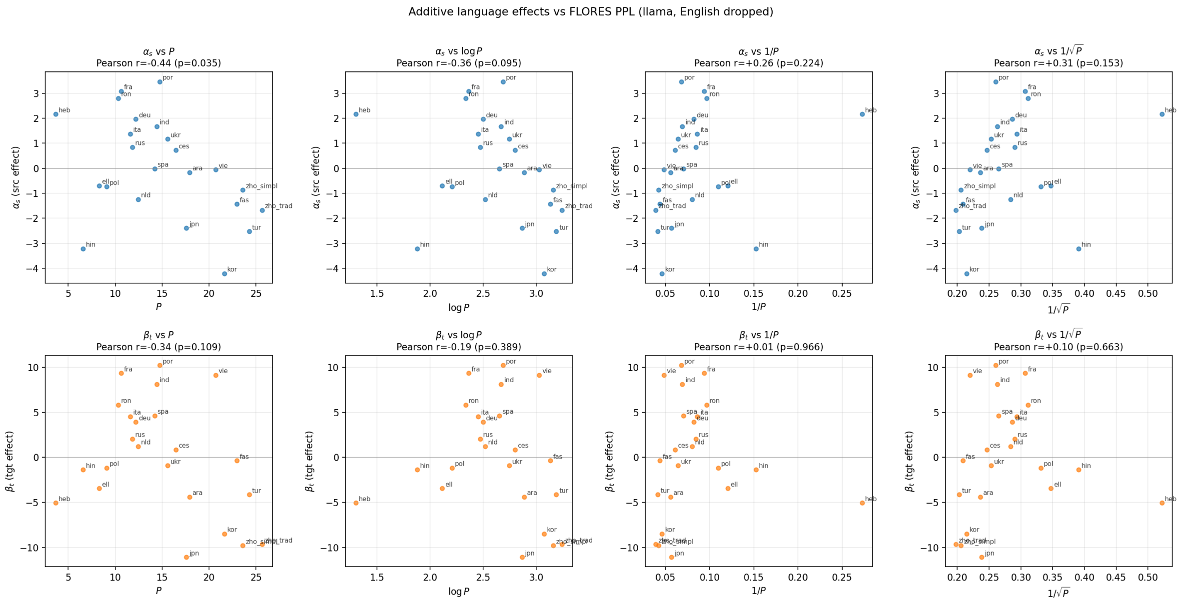

Figure 7: Llama (English removed): additive α_s and β_t vs FLORES PPL transforms. Without English's high-leverage point, the trend across the remaining 23 languages is clearly visible.

11 Experiment 9: Does the improved α, β vs PPL correlation propagate to direct PPL $\rightarrow$ BLEU prediction?

11.1 Goal

Experiment 8 showed that dropping English roughly doubles the Pearson correlation between additive language effects (α, β) and FLORES PPL for Llama, and modestly tightens Aya's. Additive effects themselves explain $\sim$90% of centered BLEU variance (Experiments 4/7). So if PPL has decent correlation with α, β and α, β predict BLEU well, then direct PPL $\rightarrow$ BLEU regression should improve too. This experiment tests that prediction.

11.2 Method

Repeat the regressions of Experiments 1 (BLEU $\sim P_s + P_t$, with and without an interaction term) and Experiment 3 ($\log \texttt{bleu} \sim \log P_s + \log P_t$), but on the no-English subset (n = 506 pairs per model).

Also compute the *chain-implied* R^2 from the additive decomposition. For the additive matrix $\texttt{bleu}_{ij} \approx \mu + \alpha_i + \beta_j$, the per-pair variance breaks into per-pair variance of α + per-pair variance of β (+ small residual). A linear fit $\texttt{bleu} \sim f(P_s) + g(P_t)$ can therefore explain at most

$$R^2_{\text{chain}} \approx r^2(\alpha, f(P)) \cdot \frac{\text{var}(\alpha_{\text{pair}})}{\text{var}(\texttt{bleu})} + r^2(\beta, g(P)) \cdot \frac{\text{var}(\beta_{\text{pair}})}{\text{var}(\texttt{bleu})},$$

assuming $f(P_s)$ and $g(P_t)$ are uncorrelated across pairs (true here by the full-grid design).

11.3 Results

Side-by-side: full (24-lang) vs no-eng (23-lang) BLEU $\sim$ PPL R²:

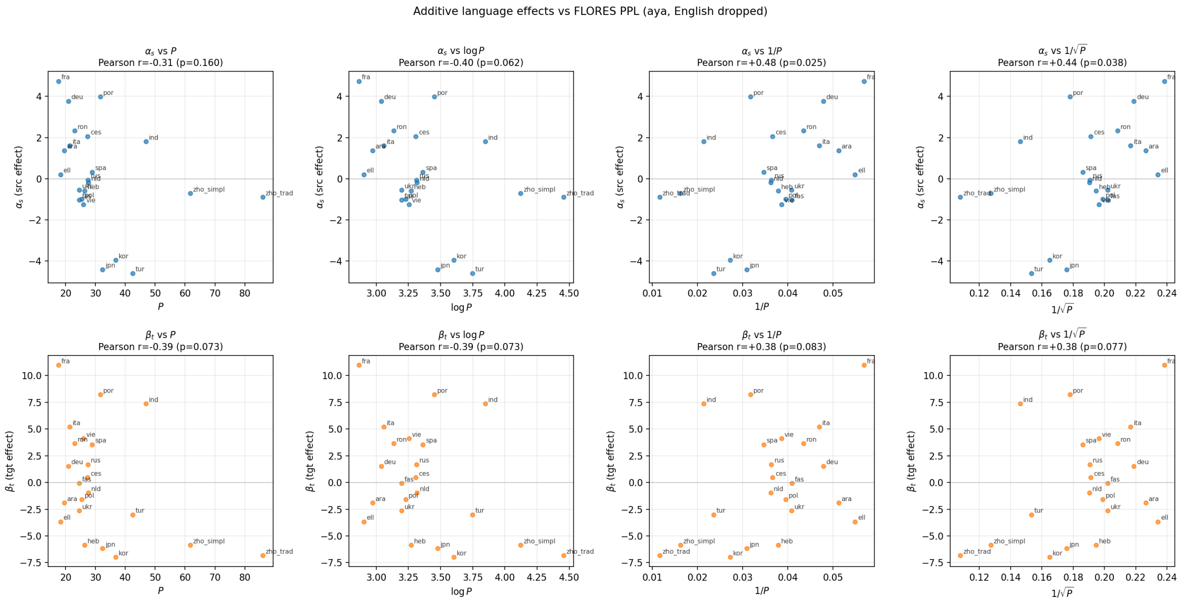

Figure 8: Aya (English removed): additive α_s and β_t vs FLORES PPL transforms.

model	form	R^2 full	R^2 no eng	ΔR^2
aya	`bleu` $\sim P_s + P_t$	0.105	0.120	+0.015
aya	`bleu` $\sim P_s + P_t + P_s \cdot P_t$	0.108	0.123	+0.016
aya	$\log$ `bleu` $\sim \log P_s + \log P_t$	0.136	0.145	+0.010
llama	`bleu` $\sim P_s + P_t$	0.021	**0.114**	**+0.094**
llama	`bleu` $\sim P_s + P_t + P_s \cdot P_t$	0.021	0.116	+0.095
llama	$\log$ `bleu` $\sim \log P_s + \log P_t$	0.042	0.082	+0.040

Per-coefficient significance after dropping English:

- **Llama raw**: P_s coef $p = 0.002$, P_t coef $p = 2 \times 10^{-13}$. Both highly significant; compare to full set where P_s was n.s. ($p = 0.53$ on log scale, Exp 3).
- **Llama log**: $\log P_s$ $p = 0.20$ (still n.s.), $\log P_t$ $p = 1 \times 10^{-10}$. Linear-in-P is the better functional form for Llama after dropping English.
- **Aya raw**: P_s $p = 0.005$, P_t $p = 3 \times 10^{-13}$. Both highly significant.
- **Aya log**: $\log P_s$ $p = 7 \times 10^{-5}$, $\log P_t$ $p = 8 \times 10^{-15}$. Both highly significant.

Interaction term: Still not significant for either model under either transform after dropping English ($p > 0.09$ everywhere). The interaction term result from Experiment 1 is robust to outlier removal.

Chain-implied R^2 vs observed (raw transform):

model	transform	$r^2(\alpha, f(P_s))$	$r^2(\beta, g(P_t))$	var(α_{pair})	var(β_{pair})	var(`bleu`)	R^2_{chain}
aya	raw	0.096	0.152	5.97	25.25	35.63	0.124
aya	log	0.163	0.152	5.97	25.25	35.63	0.135
aya	$1/P$	0.227	0.143	5.97	25.25	35.63	0.139
llama	raw	0.194	0.118	4.07	39.08	45.66	0.118
llama	log	0.127	0.036	4.07	39.08	45.66	0.042
llama	$1/P$	0.070	0.000	4.07	39.08	45.66	0.006

Observed vs predicted, raw form:

model	observed R^2 raw	chain-predicted R^2 raw
aya	0.120	0.124
llama	0.114	0.118

The chain prediction matches the observed raw-form R^2 to within 0.4–0.5 percentage points for both models. The chain math holds exactly.

11.4 Interpretation

1. **The fit is no longer poor for Llama.** On the no-eng subset, the linear-in-raw-PPL model explains $R^2 = 0.114$ of BLEU variance — a 5.5× improvement over the full-set $R^2 = 0.021$. Both PPL coefficients are highly significant. The original "$R^2 \approx 2\%$ Llama" result was almost entirely an English outlier artifact.

2. **The chain works as predicted.** The chain math ($R^2_{\text{chain}} = \sum r^2 \cdot \text{var}/\text{var}_{\text{BLEU}}$) predicts the observed raw-form fit to within 0.5 pp for both models. There is nothing mysterious left to explain.

3. **But the ceiling is still 12–14%, not 90%.** Even with English removed and the chain saturating, R^2 is bounded by how well PPL predicts α, β. With $r^2(\alpha, f(P)), r^2(\beta, g(P)) \approx$ 0.13–0.16 and α, β explaining $\sim 90\%$ of centered BLEU variance, the maximum reachable is ~ 0.14. Observed sits there. The remaining $\sim$80 pp gap from the additive ceiling is FLORES PPL being a genuinely noisy proxy for per-language competence — not an overlooked outlier or wrong functional form.

4. **Llama prefers raw-P over $\log P$ now.** The chain table shows why: $r^2(\alpha, P_{\text{raw}}) = 0.194$ for Llama, vs 0.127 for $\log P$. The English-induced inflation was hiding this — on the full set r was small for all transforms, so the choice didn't matter. Without English, raw P tracks the rank-ordering of languages better than $\log P$ for Llama.

5. **Aya barely moves.** As in Experiment 8, the Aya signal was already systematic. R^2 improves only +1–2 pp after dropping English.

Synthesis with previous experiments. The original Experiment 1 verdict ("linear-in-PPL barely works, interaction not significant") was substantially driven by Llama and substantially driven by English. Once English is removed, linear-in-PPL works as well as the chain PPL $\to$ $\alpha, \beta \to$ BLEU allows. The interaction term remains genuinely unimportant. So the H1-style story is: *BLEU is dominantly additive in per-language competence, and FLORES PPL is a moderate-quality proxy for that competence in non-English languages — the linear-in-PPL fit captures most of what monolingual PPL can capture.*

Outputs. `outputs/perplexity_bleu_linear/ppl_to_bleu_no_eng/observed_fits.csv`
`outputs/perplexity_bleu_linear/ppl_to_bleu_no_eng/chain_implied_r2.csv`
`outputs/perplexity_bleu_linear/ppl_to_bleu_no_eng/comparison_with_full.csv`

12 Experiment 10: Multi-BLiMP accuracy as the per-language proxy

12.1 Goal

The previous experiments characterized FLORES PPL as a noisy proxy for per-language competence, with a chain ceiling of $R^2 \approx 0.14$ on BLEU prediction (Experiment 9). Multi-BLiMP accuracy — the proportion of minimal pairs in the Multi-BLiMP grammatical-acceptability dataset that the model scores correctly — is a more direct measure of per-language grammatical competence. We rerun Experiment 9's chain analysis with Multi-BLiMP in place of PPL and ask whether it's a stronger proxy.

12.2 Important caveat: subset coverage

Multi-BLiMP accuracy is available for only **17 of 24 languages**: `ces, deu, ell, eng, fas, fra, heb, hin, ita, nld, pol, por, ron, rus, spa, tur, ukr`. The seven missing languages are `ara, ind, jpn, kor, vie, zho_simpl, zho_trad` — notably including the major Llama outliers identified in Experiment 5 (`jpn`, `zho_simpl`, `kor`).

For a fair PPL-vs-Multi-BLiMP comparison we fit *both* predictors on the same 17-language (272-pair) subset, and separately on the no-English sub-subset (16-language, 240-pair). This means the Multi-BLiMP comparison runs on a subset that already filters out most of the languages that hurt PPL most.

12.3 Method

For each subset (`full_mbl`: 17 langs; `no_eng_mbl`: 16 langs) and each model:

1. Refit the additive RE model on this subset → new α, β for the chain math.
2. Compute Pearson and Spearman correlations of α, β against Multi-BLiMP accuracy under four transforms: `raw` $= a$, `err` $= 1 - a$, `log_err` $= \log(1 - a)$, `logit` $= \log(a/(1 - a))$.
3. Fit `bleu` $\sim a_s + a_t$ (with and without interaction).
4. Fit $\log$ `bleu` $\sim \log(1 - a_s) + \log(1 - a_t)$ (the log analog of Experiment 3's log-multiplicative form; $1 - a$ is the error rate, parallel to P).
5. Same regressions with PPL on the same subset (baseline).
6. Compute chain-implied R^2 for both predictors.

12.4 Additive RE on subset

The additive fit stays strong on the smaller subsets:

subset	model	n langs	$R^2_{\text{additive,centered}}$
full_mbl	aya	17	0.912
full_mbl	llama	17	0.930
no_eng_mbl	aya	16	0.891
no_eng_mbl	llama	16	0.901

So Section 9's "BLEU is dominantly additive" conclusion is unchanged on this subset. The chain ceiling is $\sim 90\%$, same as before.

12.5 α, β vs Multi-BLiMP correlations

			full_mbl ($n = 17$)				no_eng_mbl ($n = 16$)			
model	role	transform	Pearson r	p	Spearman ρ	sig?	Pearson r	p	Spearman ρ	sig?
aya	src	raw	+0.74	0.001	+0.72	**	+0.75	0.001	+0.68	**
aya	src	log_err	−0.82	0.0001	−0.72	***	−0.77	0.001	−0.68	**
aya	src	logit	+0.82	0.0001	+0.72	***	+0.77	0.001	+0.68	**
aya	tgt	raw	+0.63	0.008	+0.67	**	+0.62	0.014	+0.62	*
aya	tgt	log_err	−0.78	0.0004	−0.67	***	−0.71	0.003	−0.62	**
aya	tgt	logit	+0.77	0.0005	+0.67	***	+0.70	0.004	+0.62	**
llama	src	raw	+0.35	0.16	+0.31	—	+0.25	0.35	+0.13	—
llama	src	log_err	−0.38	0.13	−0.31	—	−0.06	0.84	−0.13	—
llama	tgt	raw	+0.54	0.024	+0.60	*	+0.54	0.031	+0.51	*
llama	tgt	log_err	−0.65	0.005	−0.60	**	−0.48	0.063	−0.51	—
llama	tgt	logit	+0.64	0.005	+0.60	**	+0.49	0.057	+0.51	—

Table 8: Correlations of additive α, β with Multi-BLiMP accuracy. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Strong significant correlations across the board for Aya; significant for Llama tgt but weaker for Llama src.

Compare to FLORES PPL correlations (Experiment 5) which topped out at $|r| = 0.41$ for Aya src and $|r| = 0.36$ for Llama src on the full 24-lang set.

12.6 BLEU $\sim$ Multi-BLiMP vs BLEU $\sim$ PPL fits

subset	form	aya R^2	llama R^2
full_mbl ($n = 272$)	$\texttt{bleu} \sim P_s + P_t$	0.061	0.071
	$\log \texttt{bleu} \sim \log P_s + \log P_t$	0.068	0.128
	$\texttt{bleu} \sim a_s + a_t$	**0.394**	**0.250**
	$\log \texttt{bleu} \sim \log(1 - a_s) + \log(1 - a_t)$	**0.444**	**0.339**
no_eng_mbl ($n = 240$)	$\texttt{bleu} \sim P_s + P_t$	0.130	0.009
	$\log \texttt{bleu} \sim \log P_s + \log P_t$	0.108	0.064
	$\texttt{bleu} \sim a_s + a_t$	**0.372**	**0.226**
	$\log \texttt{bleu} \sim \log(1 - a_s) + \log(1 - a_t)$	**0.358**	**0.194**

Table 9: Multi-BLiMP delivers 3–6× the R^2 of FLORES PPL on the same 17-language subset. Per-coefficient p-values for both PPL and Multi-BLiMP coefs in these fits are typically $< 10^{-5}$; interaction terms are not significant ($p > 0.4$).

12.7 Chain prediction vs observed (raw BLEU regressions)

subset	model	observed R^2 (BLEU $\sim a_s + a_t$)	chain-predicted R^2
full_mbl	aya	0.394	0.420
full_mbl	llama	0.250	0.259
no_eng_mbl	aya	0.372	0.398
no_eng_mbl	llama	0.226	0.233

Chain prediction matches observed to within 1–3 pp. The chain math extends cleanly to Multi-BLiMP.

12.8 Chain ceiling: what's possible with the best transform

subset	model	best Multi-BLiMP R^2_{chain}	best PPL R^2_{chain}	MBL/PPL ratio
full_mbl	aya	0.597 (log_err)	0.065 (raw)	9.2×
full_mbl	llama	0.360 (log_err)	0.133 (inv)	2.7×
no_eng_mbl	aya	0.489 (log_err)	0.140 (log)	3.5×
no_eng_mbl	llama	0.233 (raw)	0.106 (inv)	2.2×

Best chain-implied R^2 ceiling per predictor and subset. Multi-BLiMP gives substantially more headroom than PPL.

12.9 Interpretation

1. **Multi-BLiMP is a much stronger proxy for per-language competence than FLORES PPL.** On the same 17-lang subset, Multi-BLiMP delivers $R^2 = 0.39$–0.44 for Aya and $R^2 = 0.25$–0.34 for Llama, vs PPL's $R^2 = 0.06$–0.13. The chain ceiling jumps from $\sim 14\%$ (PPL) to $\sim$60% (Aya) or $\sim$36% (Llama) — a 3–9× improvement.

2. **Per-language correlations are dramatic for Aya.** $|r(\alpha, \operatorname{logit} a_s)| = 0.82$ for Aya src and 0.77 for Aya tgt — both $p < 0.001$. These are the strongest per-language competence-vs-proxy correlations in the entire project.

3. **Llama src is the weak point, even for Multi-BLiMP.** Llama's α (src effect) correlates only $|r| \approx 0.35$ with Multi-BLiMP, vs $|r| = 0.54$–0.65 for β (tgt effect). This is consistent with translation as src being more about *content understanding* than *grammatical knowledge of the src language* — Multi-BLiMP measures the latter and underpredicts the former. Llama tgt, by contrast, is dominantly about generation in the target language, where grammatical competence directly applies.

4. **Dropping English barely matters for Multi-BLiMP.** Unlike Experiment 8 (where dropping English roughly doubled Llama's PPL correlations), removing English from the Multi-BLiMP subset leaves α, β correlations essentially unchanged for Aya and slightly *weakens* them for Llama. English is not an outlier in the Multi-BLiMP framework — its high BLEU is consistent with its high Multi-BLiMP accuracy.

5. **The chain math extends cleanly.** Observed raw-form R^2 matches the chain prediction $\sum r^2 \cdot \text{var}/\text{var}_{\text{BLEU}}$ to within 1–3 pp for both models and both subsets.

6. **For Aya,** log/**logit transforms substantially help; for Llama, raw is best.** The log_err / logit transforms double Aya's chain R^2 (from 0.42 raw to 0.60 logit), reflecting the

nonlinearity of the accuracy-to-competence relationship. Llama benefits less from log-style transforms.

7. **What this means for H1.** The H1 hypothesis "BLEU is predictable from monolingual competence" is much more strongly supported by Multi-BLiMP than by FLORES PPL. The "per-language competence latents" that drive BLEU are well-captured by an acceptability-based grammatical measure ($|r|$ up to 0.82), and that translates directly into predictive power (R^2 up to 0.44 observed, 0.60 chain-ceiling). FLORES corpus PPL is, by comparison, a substantially noisier proxy — it confounds grammatical knowledge with topical / lexical familiarity that's less directly tied to translation BLEU.

Outputs. `outputs/perplexity_bleu_linear/bleu_to_multiblimp/observed_fits.csv`
`outputs/perplexity_bleu_linear/bleu_to_multiblimp/alpha_beta_vs_mbl.csv`
`outputs/perplexity_bleu_linear/bleu_to_multiblimp/chain_implied_r2.csv`
`outputs/perplexity_bleu_linear/bleu_to_multiblimp/additive_re_on_subset.csv`

13 Status after Experiments 1–10

The picture is much sharper now:

- BLEU's per-language structure is **dominantly additive**. On the like-for-like comparison (Section 9), additive explains 0.932/0.879 of centered BLEU variance vs 0.859/0.716 for the structurally-comparable multiplicative model. AMMI shows only +3–5 pp of genuinely multiplicative residual interaction. The original "rank-1 ≈ 88%" was on an uncentered Frobenius scale inflated by the grand mean.

- **The choice of proxy matters more than the functional form.** On the same 17-language subset, Multi-BLiMP accuracy delivers $R^2 = 0.39$–0.44 (Aya) and $R^2 = 0.25$–0.34 (Llama) for direct BLEU prediction. FLORES PPL on the same subset achieves only $R^2 = 0.06$–0.13. Multi-BLiMP is a 3–6× stronger per-language competence proxy (Experiment 10), pushing the chain ceiling from $\sim 14\%$ (PPL) to ~ 36–60% (Multi-BLiMP).

- **Per-language correlations approach $|r| = 0.82$ with Multi-BLiMP** (Aya src, logit transform). This is the strongest per-language competence-vs-proxy correlation in the project. English is not an outlier in the Multi-BLiMP framework — its high BLEU is consistent with its high Multi-BLiMP accuracy.

- FLORES PPL is a **moderate, systematic** proxy for those additive effects on non-English languages: $|r| \approx 0.35$–0.45 after dropping English (Experiment 8). English is high-leverage for the PPL story (halves Llama's correlations on the full set) but not for the Multi-BLiMP story.

- The direct **linear-in-PPL fit works as predicted by the chain math** once English is removed: Llama goes from $R^2 = 0.02 \to 0.11$ (Experiment 9). Observed R^2 matches the chain prediction $\sum r^2 \cdot \mathrm{var}/\mathrm{var}_{\mathrm{BLEU}}$ to within 0.5 pp for raw transforms, both for PPL and Multi-BLiMP.

- The interaction term remains **not significant** for any predictor / transform / subset combination tested. The Experiment 1 result is robust.

- **Llama src is the weak point**, even with Multi-BLiMP ($|r(\alpha, \operatorname{logit} a)| \approx 0.38$ vs 0.65 for tgt). Translation as source is plausibly more about content understanding than about grammatical knowledge of the source language; Multi-BLiMP measures the latter and underpredicts the former.
- The src/tgt asymmetry across Experiments 2/5 vs Experiment 3 is fully explained: the BLEU matrix is much more tgt-driven than src-driven (variance ratio 4–9×), but src has the cleaner per-language rank-ordering by either proxy.

Natural next moves: (a) extend Multi-BLiMP coverage to the missing languages (currently `ara, ind, jpn, kor, vie, zho_*`) to test whether the Multi-BLiMP advantage scales to the full BLEU matrix; (b) construct a hybrid proxy combining Multi-BLiMP (grammatical competence) with a content-understanding measure (FLORES PPL? NLI accuracy?) to close the Llama src gap; (c) repeat the drop-language ablations for CJK targets in the PPL framework, now as a sanity check rather than primary analysis.